

Image Processing Methods for Detecting Voids in the Solder Joints of Surface-Mounted Components

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Abstract: Void formation in the lead-free solder joints of electrical components can significantly affect their mechanical properties and thermal performance. Void-detection methods have been investigated in this paper, which are able to identify and measure voids in meniscus-shaped solder joints of surface mounted components automatically. In the experiment, 0603 size resistors (1.5×0.75 mm) were soldered onto a testboard, and X-ray images were acquired about the samples then. The image processing methods for the void detection were implemented in OpenCV-Python. At first, the properties of the images were analyzed (for example, calculating the histogram), and then the area, including the resistor and solder pads, was extracted by masking. Several methods were investigated for identifying voids in the images; Canny edge detection, global- and adaptive thresholding, and blob detection algorithm. The accuracy of the extraction methods was evaluated by identifying voids on a test image manually and comparing the results to that provided by the different methods. Canny edge detection was the best in our case, but global thresholding and blob detection are also promising solutions.

1. INTRODUCTION

Reflow soldering is the widespread technology for establishing mechanical and electrical connections between the electrical components (mostly surface-mounted ones) and the printed circuit board [1,2]. Since the transition to the lead-free soldering technologies, void formation within the solder joints – primarily via gas formation by the solvents of the flux – has become a recurring failure [3]. The void formation is facilitated by the higher surface tension of lead-free solder alloys in liquid form (for lead-free, it is around 550 mN/m, for lead-bearing, it is around 480 mN/m [4]), making the escape of the voids more difficult. Since void formation decreases the mechanical strength of solder joints and lowers their thermal performance [5], inspecting this failure during manufacturing is crucial, which can basically be carried out by X-ray inspection machines.

Voids can be analyzed in the solder joints typically by two types of X-ray inspection principles, Computed Tomography (CT) and Transmissive X-ray. Computed Tomography can reveal the voids with high certainty, and inspection is possible with high accuracy [6, 7].

Additional features like virtual cross-sectioning are also available. Though, this method is slow (numerous images shall be acquired from different angles, which shall be combined into a 3D voxel representation of the solids then), rendering its usage in a manufacturing environment less feasible. On the other hand, transmissive X-ray machines rapidly provide information about the product under investigation, and voids can automatically be measured by image processing methods [8]. The accuracy of these measurements depends, though, on the goodness of image segmentation, that is, extracting the area of voids from the area of solder joints. In the case of simpler joint geometries (like in the case of BGA joints), the image segmentation can be performed easier (e.g., by simple thresholding with a given value), yielding an accurate determination of the void area [9]. However, in the case of more complex solder joint geometries (e.g., meniscus-shaped joints of SM resistors), the image segmentation is much more difficult because of the luminance gradients within the joint area in the X-ray image. Therefore, we aimed to investigate different image processing techniques for automated extraction of voids area in complex joint geometries.

2. MATERIALS AND METHODS

2.1. Experimental for preparing solder joints

To prepare solder joints, 0603 ($1.5 \times 0.75 \text{ mm}^2$) size surface mounted resistors were soldered onto an FR-4 based testboard with vapor phase soldering. The testboard was 1.55 mm thick, and the surface finish was immersion silver. The lead-free solder paste (SAC305 – Sn96.5Ag3Cu0.5) was deposited with stencil printing using a 150 μm thick, laser-cut, stainless steel foil. The soldering was performed using an Asscon VP800 machine without a vacuum option, and the working fluid was Galden LS230. The parameters of the vapor phase soldering were set to obtain a heating factor of about 1000 $^\circ\text{C}\cdot\text{s}$. The maximum temperature was 233 $^\circ\text{C}$ and the time above liquidus was 96 seconds (Fig. 1).

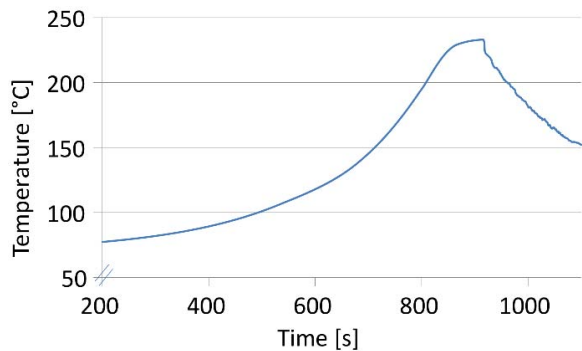


Fig. 1. Thermal profile for vapor phase soldering.

2.2. Image processing for void detection

Image processing methods were implemented in an OpenCV-Phyton environment to detect the voids in meniscus-shaped solder joints. The X-ray image of the resistor has been cropped at first to keep only the portion of interest (solder pads, solder joints, resistor itself), which is illustrated in Fig. 2.

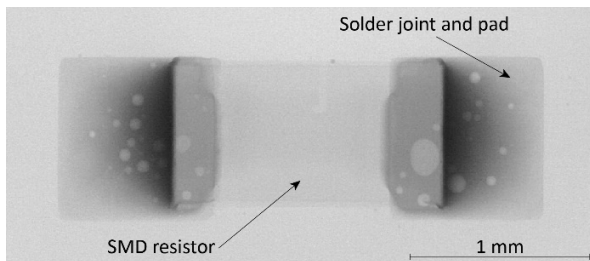


Fig. 2. The area under investigation: SMD resistor, solder pads and solder joints.

The area, including the surface-mounted resistor and the belonging solder pads, was extracted at first from

the background by masking. Since the background in the original image is noisy, Gauss filtering was used afore to eliminate the noise as much as possible. Then the mask for extracting the features was created based on segmenting the original image with thresholding, which means selecting an intensity value and setting all pixels darker than that to 0 or 1, and lighter than that to the opposite 1 or 0, respectively (Fig. 3b). The optimal value for thresholding was determined based on the histogram of the image (Fig. 3a). The mask was used then to reveal all the areas belonging to the component and hide any other features in the background.

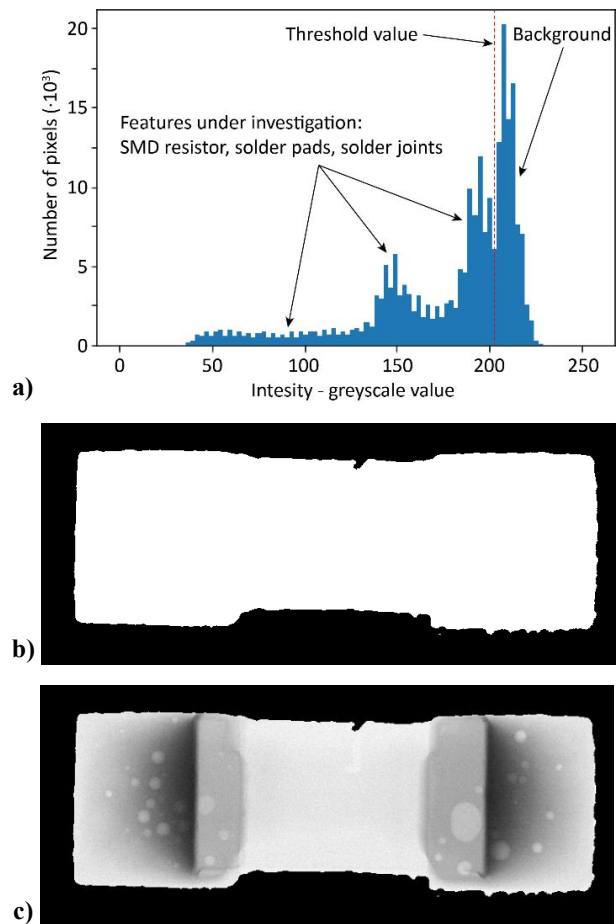


Fig. 3. Separating the area of the SMD resistor, pads and solder joints from the background: **a)** image histogram for determining the optimal thresholding value for segmenting and mask creation; **b)** mask (white reveals, black hides); **c)** the extracted area by masking.

Several methods were investigated for identifying the voids within the solder joints. This included using the Canny edge detection algorithm, global thresholding and labelling the area of voids, adaptive thresholding including image closing with dilation and

erosion, and blob detection algorithm. The goodness of the different methods has been characterized by comparing the number of identified voids to that counted manually.

3. RESULTS

The Canny edge detection algorithm uses derivative filters to compute the horizontal and vertical derivatives of the image. By the two directional derivatives, it determines the direction of maximum intensity change, i.e., the image gradient. Passing through each edge-like point, the gradient is used to determine the points with the largest derivative value. These are retained, and the other points are moved to 0. The resulting image by the Canny algorithm is illustrated in Fig. 4. The number of identified voids was about 20.



Fig. 4. Voids identified by the Canny edge detection algorithm; the number of identified voids is around 20.

The second method included segmenting the image with different global threshold values separately, labelling the voids in each segmented image, and merging the individual results. Using a single threshold would not be sufficient since the solder joint has different thicknesses that will show a difference in luminosity in the image. The number of identified voids was 14 (Fig. 5), but the identification can further be enhanced by using more distinct thresholding values. Though, this might be a resource-intensive approach.

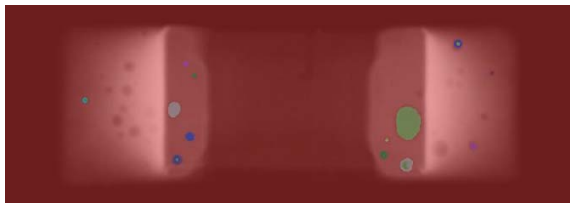


Fig. 5. Voids identified by global thresholding and labelling; the number of identified voids is around 14.

Thirdly, adaptive thresholding was used to identify the voids. The idea behind it is that the threshold value is calculated based on the local environment of a pixel, so the thresholding result may be different for each

pixel. In order to better highlight the boundaries of the voids, additional image operations were exploited, namely the "closing" operation and the "dilatation" operation. Although the resulting image is noisy, the noise can be removed by setting a size limit for image features to be excluded. The number of identified voids with this method is around 12.

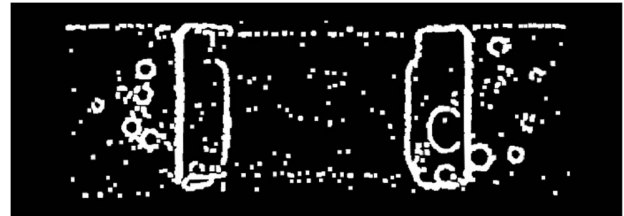
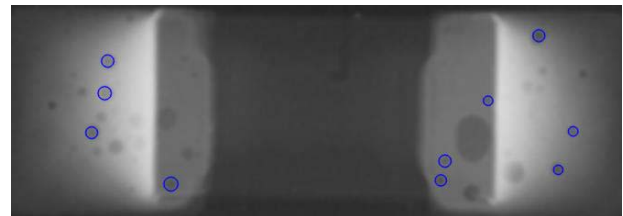
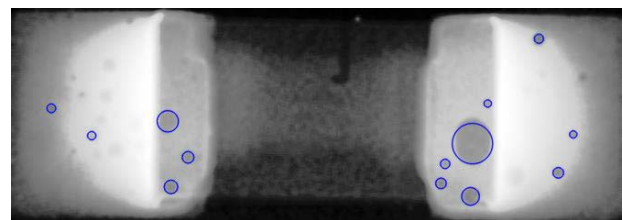


Fig. 6. Voids identified by adaptive thresholding; the number of identified voids is around 12.

Lastly, the blob detection algorithm was investigated, which allows setting different filtering parameters for detecting voids. Running the algorithm on the simply inverted and noise-filtered image showed relatively good results, identifying 10 voids (Fig. 7a). Secondly, histogram smoothing was applied to the image, and 12 voids were identified (Fig. 7b). Combining the results of the two, a total of 15 voids were detected.



a)



b)

Fig. 7. Voids identified by the blob-detection algorithm: a) on the simply inverted and noise-filtered image (10 voids); b) histogram-smoothed image (12 voids).

Lastly, the voids in the solder joints were counted manually to assess the performance of the different void detection algorithms. Counting the voids manually in

the test image resulted in 40 voids (Fig. 8). The comparison of the manual count to the void detection methods is illustrated in Fig. 9.

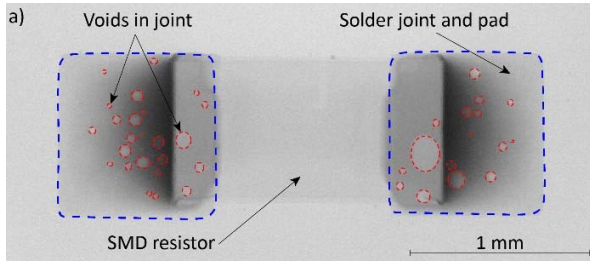


Fig. 8. Manual count of voids, number of voids is 40.

It should be noted that such small voids can be found manually, whose identification with an automated image processing method might not be impossible. Though calculating the ratio of total void area to the joint area, excluding the small area voids by the automated method yields an insignificant error.

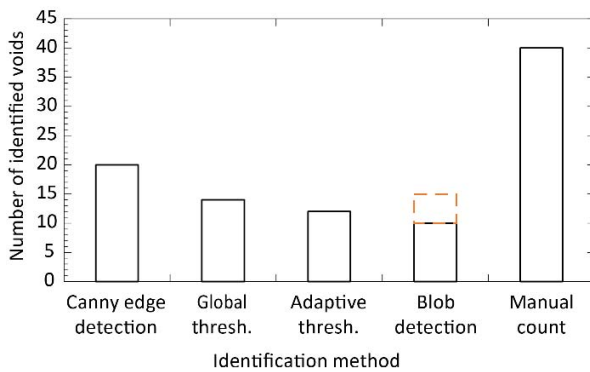


Fig. 9. The number of voids identified by the different methods (the dashed line at blob detection indicates the number by merging detection results with multiple settings).

The number of the identified voids was the largest in the case of the Canny edge detection algorithm. Additionally, global thresholding and blob detection by using several more distinct thresholding and filtering parameters and merging results are promising solutions.

4. CONCLUSION

Detection methods were investigated for analyzing void formation in lead-free solder joints with complex geometries, like the meniscus-shaped joints of surface mounted resistors. The investigated methods can provide even better results if multiple thresholding and/or filtering parameters are utilized, and the results are merged together. Still can happen, though, that such

small voids, which can be found manually, cannot be identified with an automated image processing method. Though, when calculating the ratio of total void area to the joint area, excluding the small area voids by the automated method yields an insignificant error. Continuation of the work will include optimizing the promising methods (Canny edge detection, global thresholding and blob-detection) for better void identification, calculating the void area automatically and determining the calculation error.

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